HYBRID AG
PRECISION NUTRITION

MYCRO-TEC ®

DIGESTER

AVAILABLE IN



DIGEST · CYCLE · RETURN

ALL PROGRAMS

Residue isn't a problem to burn through — it's a nutrient bank waiting to reopen.

01

DIGEST

Enzymes open cellulose and complex plant residue



02

CYCLE

Residue-bound nutrients move back into the soil cycle



03

RETURN

Returned carbon and nutrients support the next crop



AFTER HARVEST

PRODUCT	AMM. SULPHATE
1–2 L/ha	8 kg/ha

BOTH IN 100 L/ha WATER

Cereal and legume stubble.

HEAVY RESIDUE

PRODUCT	AMM. SULPHATE
up to 3 L/ha	8 kg/ha

BOTH IN 100 L/ha WATER

Canola and heavy loads.

RESIDUE PRESENT

PRODUCT	AMM. SULPHATE
1–2 L/ha	8 kg/ha

BOTH IN 100 L/ha WATER

Orchard and crop residue. Not on nut trees.

INSIDE THIS BROCHURE

PAGE 2 Where Digester fits

PAGE 3 Where Digester sits

PAGE 4 Composition & function

PAGE 5 Mode of action

PAGE 6 Better together

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Where Digester fits.

The living residue layer that resets the surface for the next productive cycle.

OPTIMISING INPUTS · WHY DIGESTER FITS

Crop residue is a resource waiting to be cycled.

Digester places four screened organisms across crop residue — using enzyme activity to break down cellulose and lignin, cycle nutrients and rebuild the biological surface layer.



ROLE	APPLICATION	WINDOW
residue-digestion layer	*broadcast across residue*	*post-harvest, into moisture*

THE RESIDUE IS STILL THERE

Stubble that holds into the next crop window ties up nutrition and ground.

WHAT DIGESTER CHANGES

Selected decomposers begin structural breakdown at the moist residue base.

NITROGEN ALONE ISN'T DIGESTION

Nitrogen feeds whatever biology is resident. It does not add decomposers.

WHAT DIGESTER CHANGES

Digester adds the decomposers; the 8 kg/ha ammonium sulphate stays a starter.

MOISTURE IS THE WINDOW

Breakdown runs on residue moisture. Prolonged hot and dry stalls it.

WHAT DIGESTER CHANGES

Timed to forecast rain after harvest, it works while the stubble holds damp.

WHAT HAS TO BE TRUE

Digester works on labelled crop residue into a viable moisture window, with the approved recipe — nothing in-crop, nothing on nut trees.

MAXIMISING OUTCOMES · WHAT GROWERS SEE

Last season's straw is next season's nutrition, still locked in the residue.

BREAKS DOWN RESIDUE

Screened fungi and bacteria colonise stubble and begin enzymatic digestion.

PROGRAM SIGNAL · RESIDUE LOSS



CYCLES NUTRIENTS BACK

Nutrients held in residue move back toward the biologically active soil layer.

PROGRAM SIGNAL · NUTRIENT RETURN



BUILDS SOIL STRUCTURE

Decomposed residue contributes organic matter and supports better aggregation.

PROGRAM SIGNAL · SOIL CONDITION



HOW YOU CHECK IT

Digestion starts at the stubble base where moisture sits — treated stubble pulls away more easily as it runs.

PROGRAM SIGNAL · PULL TEST



Digest the residue. Cycle the nutrients. Reset the biological surface.

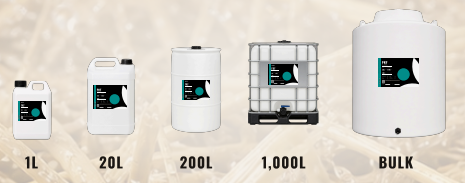
PRECISION NUTRITION
Physical · Mineral · Biological

OPTIMISING INPUTS
Addressed · Optimised · Enabled

MAXIMISING OUTCOMES
Measured · Proven · Delivered

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Where Digester sits in your program.

The same four lanes your program already runs.

THE SEASON · FIVE WINDOWS, ONE PROGRAM

Simplified, illustrative program to show you when it goes on — actual programs vary with crop, variety, soil, season, test data, goals.



POST-HARVEST

GROUND SPREAD
—

SOIL PRIMER
—

FOLIAR
Broadcast on residue

+ Digester + AMS 8 kg

FERTIGATION
—



ESTABLISHMENT

GROUND SPREAD
Prescription dry blend

SOIL PRIMER
—

FOLIAR
—

FERTIGATION
—



EARLY GROWTH

GROUND SPREAD
—

SOIL PRIMER
—

FOLIAR
Trace elements

FERTIGATION
Base nutrition



FLOWERING & SET

GROUND SPREAD
—

SOIL PRIMER
—

FOLIAR
Calcium · boron

FERTIGATION
Base nutrition



FRUIT FILL

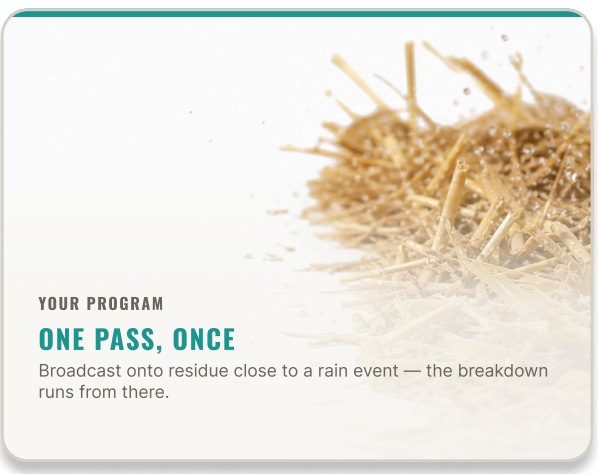
GROUND SPREAD
—

SOIL PRIMER
—

FOLIAR
Calcium · potassium

FERTIGATION
Base nutrition

One pass, once — 1–2 L/ha with the 8 kg starter, close to rain. The season runs while the stubble breaks down.



YOUR PROGRAM
ONE PASS, ONCE
Broadcast onto residue close to a rain event — the breakdown runs from there.



NEXT · PAGE 4
COMPOSITION & FUNCTION
What Digester actually is — the organisms inside it and the job each one does.



THEN · PAGE 5
MODE OF ACTION
The stage-by-stage breakdown for agronomists and technical growers.

ALL OUR PROGRAMS CONTAIN

NUTRIENT LOADING
Every product's N, P, K, S, Ca, Mg and trace contribution, totalled per hectare.

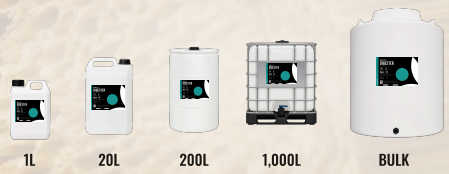
COST PER HECTARE
Each pass costed per hectare — judged on what it costs to grow the crop.

AGAINST THE TARGET
Supply weighed against target yield and removal, so the program balances.

Build the biology in. Carry the response through the season.

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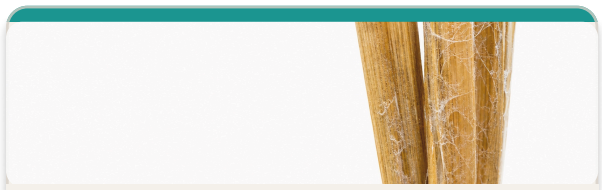
AVAILABLE IN



Composition & function.

Four screened organisms selected to digest cellulose, hemicellulose, lignin and pectin in crop residue.

INSIDE DIGESTER · WHAT EACH GROUP DOES



CELLULOSE DEGRADERS

WHAT IT IS

Trichoderma harzianum and Aspergillus niger — high-activity fungal decomposers.

WHAT IT DOES

Produce enzyme systems that open cellulose-rich crop residue and accelerate early breakdown.

CONTAINS

T. harzianum >2×10⁸ · A. niger >2×10⁷

EARLY RESIDUE OPENING



COMPLEX-RESIDUE TEAM

WHAT IT IS

Streptomyces lydicus and Penicillium radicum — complementary residue colonisers.

WHAT IT DOES

Work across tougher structural material, extend the digestion zone and support nutrient return.

CONTAINS

S. lydicus >2×10⁷ · P. radicum >1×10⁷

LIGNIN + COMPLEX RESIDUE



ENZYME & CYCLING ENGINE

WHAT IT IS

The four-organism consortium working across the residue surface.

WHAT IT DOES

Colonises stubble, produces degradative enzymes and returns residue-bound carbon and nutrients to the soil cycle.

CONTAINS

Cellulose · lignin · complex residue

DIGESTION + NUTRIENT RETURN

TYPICAL ANALYSIS · VIABLE ORGANISMS

Trichoderma harzianum	Aspergillus niger	Streptomyces lydicus	Penicillium radicum
T. harzianum	*A. niger*	*S. lydicus*	*P. radicum*
>2×10⁸	>2×10⁷	>2×10⁷	>1×10⁷
CFU/mL	*CFU/mL*	*CFU/mL*	*CFU/mL*

Concentrations at production. Formula per the current Digester label; use the final label and SDS for directions.

MIXING · STORAGE · SAFETY

MIXING & USAGE

Shake well. Add 8 kg/ha ammonium sulphate, then Digester, into 100 L/ha non-chlorinated water. Strain as you pour; 150 µm minimum nozzles and filters. Apply soon after mixing.

STORAGE & HANDLING

Store at 5–40 °C, out of direct sunlight, drums upright — lids are vented. Do not freeze. 12-month shelf life; do not keep opened drums.

SAFETY

Keep out of reach of children. May irritate skin or eyes. Follow the current label and SDS.

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Mode of action.

Digester works in five linked steps — from residue placement to nutrient return.

THE FIVE-STAGE CASCADE

Designed for the agronomist who needs to see how standing residue becomes active soil resource.

1 APPLICATION & PLACEMENT

Broadcast across crop residue in non-chlorinated water with ammonium sulphate, timed close to rain or moisture.

KEY · EVEN RESIDUE COVERAGE



2 RESIDUE COLONISATION

The four screened organisms establish first at the moisture-holding residue–soil interface, then expand through moist stubble.

KEY · LIVING BIOLOGY ESTABLISHED



3 ENZYME PRODUCTION

The consortium produces degradative enzymes that open cellulose, hemicellulose, lignin and pectin in crop residue.

KEY · CELLULOSE + LIGNIN ACTIVITY



4 RESIDUE BREAKDOWN

Structural residue softens and fragments as the biological digestion zone expands.

KEY · CARBON STRUCTURE OPENED



5 NUTRIENT RETURN

Residue-bound nutrients and carbon return to the active soil cycle, supporting the next crop.

KEY · NEXT-CROP RESOURCE



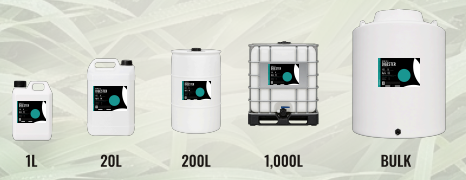
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Better together.

Three complementary biological jobs — digest the residue, unlock tied phosphorus, colonise the living zone.

DIGESTER + SP + LOCKOUT · THREE COMPLEMENTARY JOBS



DIGESTER · DIGESTS RESIDUE

ROLE

Cellulose- and lignin-degrading microbes and their enzymes break down last season's stubble.

PAYOFF

Returns the nutrition tied up in crop residue to the soil, and improves structure and water-holding for the next crop.

DIGEST
Breaks down residue

MYCRO-TEC DIGESTER



SP · UNLOCKS NUTRIENT

ROLE

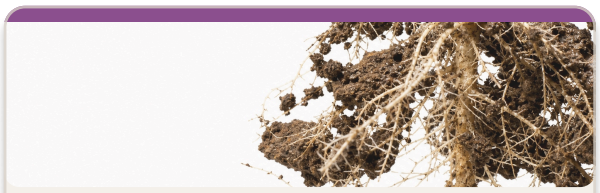
Four screened microbes secrete gluconic, oxalic and citric acids that break the bonds locking up soil nutrient.

PAYOFF

Helps release phosphorus fixed with calcium, aluminium and iron.

UNLOCK
Releases fixed P

MYCRO-TEC SP



LOCKOUT · COLONISES THE ZONE

ROLE

Purpose-selected living biology colonises the rhizosphere and competes for space and food.

PAYOFF

Supports a more balanced root-zone microbiome and a productive living zone.

COLONISE
Colonises the living zone

MYCRO-TEC LOCKOUT

PROUD PARTNERSHIP WITH GREAT SOUTHERN BIOLOGY · AUSTRALIAN-MADE

Digester, Mycro-Tec SP and Mycro-Tec Lockout are live, Australian-made biology developed with Great Southern Biology.

THE PROGRAM IN PRACTICE

STEP 1 · DIGEST

Spray Mycro-Tec Digester onto crop residue with ammonium sulphate, close to a rain event, to break the stubble down.



STEP 2 · UNLOCK

Apply Mycro-Tec SP to soil, on fertiliser or through fertigation to free locked phosphorus and minerals.



STEP 3 · COLONISE

Apply Mycro-Tec Lockout ahead of the window to colonise the root zone and compete for space and food.



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